



ASSIGNMENT No: 9

TITLE: Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

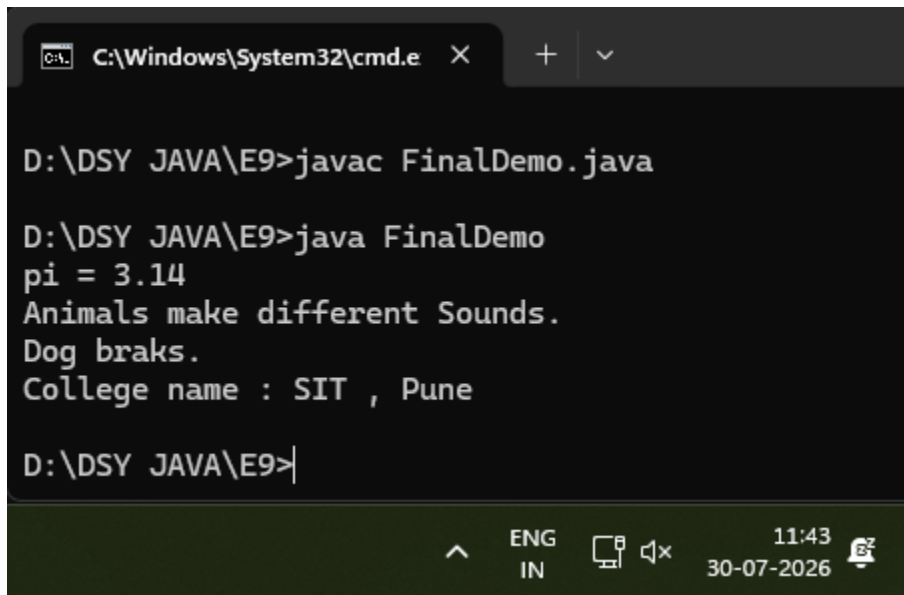
The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

Code

```
final class College {  
  
    void display(){  
  
        System.out.println("College name : SIT , Pune");  
  
    }  
}  
class Animal{  
    final void Sound(){  
        System.out.println("Animals make different Sounds.");  
    }  
}  
  
class Dog extends Animal{  
  
    void displayDog (){  
        System.out.println("Dog braks.");  
    }  
  
    // void Sound(){  
    //     System.out.println("Dog Sound.....");  
}
```

```
    // }  
}  
  
public class FinalDemo{  
  
    public static void main(String a[]){  
        final double pi = 3.14;  
  
        System.out.println("pi = "+pi);  
  
        // pi =5.20;  
  
        Dog d = new Dog();  
        d.Sound();  
        d.displayDog();  
  
        College c = new College();  
        c.display();  
  
    }  
}
```

Output



```
C:\Windows\System32\cmd.e X + v  
  
D:\DSY JAVA\E9>javac FinalDemo.java  
  
D:\DSY JAVA\E9>java FinalDemo  
pi = 3.14  
Animals make different Sounds.  
Dog braks.  
College name : SIT , Pune  
  
D:\DSY JAVA\E9>|  
  
^ ENG 11:43  
IN 30-07-2026
```

Exercise

1. Create a **Bank Account program** where account number is final and cannot be changed.

Code

```
class BankAccount {
    final int accountNumber;
    String name;
    double balance;

    BankAccount(int accNum, String holderName, double startBalance) {
        accountNumber = accNum;
        name = holderName;
        balance = startBalance;
    }

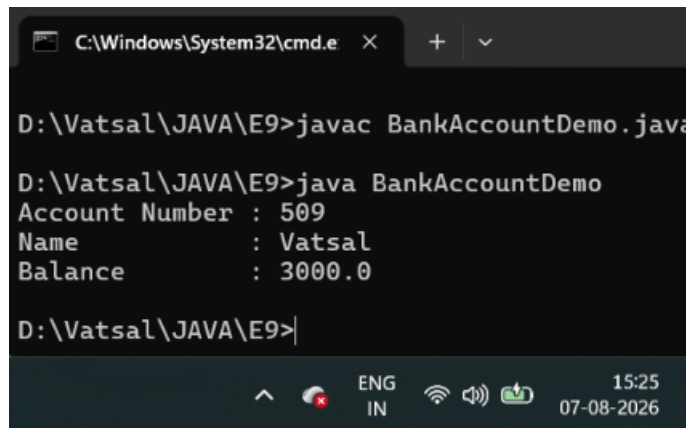
    void display() {
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Name          : " + name);
        System.out.println("Balance       : " + balance);
    }
}

public class BankAccountDemo {
    public static void main(String[] args) {

        BankAccount acc = new BankAccount(509, "Vatsal", 3000.0);
        acc.display();

    }
}
```

Output



```
C:\Windows\System32\cmd.e X + v

D:\Vatsal\JAVA\E9>javac BankAccountDemo.java

D:\Vatsal\JAVA\E9>java BankAccountDemo
Account Number : 509
Name          : Vatsal
Balance       : 3000.0

D:\Vatsal\JAVA\E9>|

15:25
07-08-2026
```

2. Create a **Library Book Management** program where the **book ISBN** is declared as **final** and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

Code

```
class LibraryBook {

    final String isbn;
    String title;
    String author;
    double price;

    LibraryBook(String bookIsbn, String bookTitle, String bookAuthor, double bookPrice) {
        isbn = bookIsbn;
        title = bookTitle;
        author = bookAuthor;
        price = bookPrice;
    }

    void display() {
        System.out.println("ISBN : " + isbn);
        System.out.println("Title : " + title);
        System.out.println("Author : " + author);
        System.out.println("Price : $" + price);
    }
}

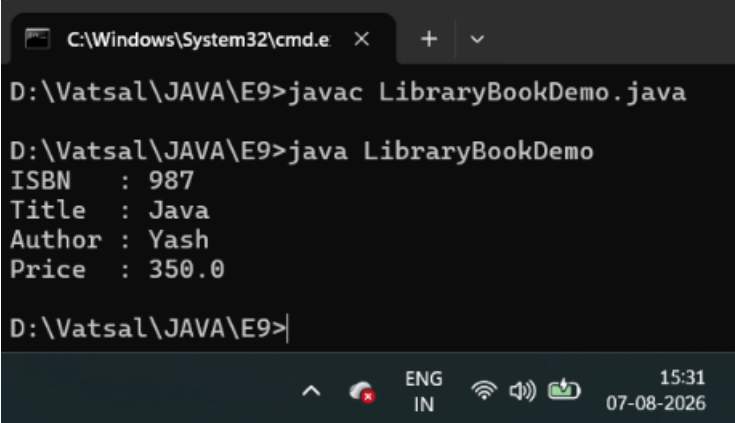
public class LibraryBookDemo {
    public static void main(String[] args) {

        LibraryBook book = new LibraryBook("987", "Java", "Yash", 45.00);

        book.display();

        // Attempting to change the ISBN
        // book.isbn = "123";
        // ^ ERROR: cannot assign a value to final variable isbn
    }
}
```

Output



```
C:\Windows\System32\cmd.e × + v
D:\Vatsal\JAVA\E9>javac LibraryBookDemo.java
D:\Vatsal\JAVA\E9>java LibraryBookDemo
ISBN : 987
Title : Java
Author : Yash
Price : 350.0
D:\Vatsal\JAVA\E9>
```

The screenshot shows a Windows command prompt window with the title bar 'C:\Windows\System32\cmd.e'. The command prompt is at the directory 'D:\Vatsal\JAVA\E9'. The user has entered 'javac LibraryBookDemo.java' to compile the program. Then, they entered 'java LibraryBookDemo' to run it. The output of the program is displayed as follows:

- ISBN : 987
- Title : Java
- Author : Yash
- Price : 350.0

The command prompt shows the prompt 'D:\Vatsal\JAVA\E9>' again, indicating the program has finished execution. The taskbar at the bottom shows the system clock as 15:31 on 07-08-2026, along with icons for network, volume, and other system utilities.